How a Hand Works

01 What keeps a finger stiff?



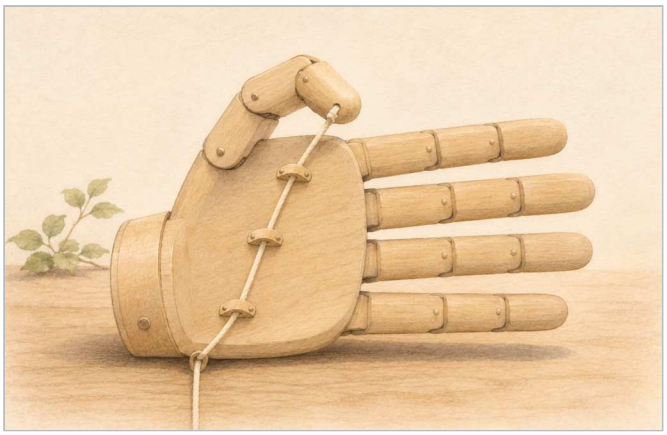
Stiff pieces in a row, like the bones in a finger.

Look closer. Find the short stiff pieces in each toy finger, and count them.

A little explanation. Bones. Each finger has three small bones in a row, and the thumb has two. Counting the wrist and palm, one hand has 27 bones. Bones hold the shape, but they cannot move by themselves.

Try this. Gently press along one of your fingers and count the hard parts between the bends. Count the stiff pieces on one toy finger too. Do the two numbers match?

02 What pulls a finger closed?



Pull a string, and one finger curls.

Look closer. Find the thin string that runs along the palm side of a curled toy finger.

A little explanation. In the toy, a string. In your hand, a tendon: a tough cord that joins a muscle to a bone. Muscles in your forearm pull long tendons that reach into each finger, and tough loops along the finger hold them close, like the toy's guides.

Try this. Draw around your hand on paper to make an outline. Pull one toy string slowly and watch which finger curls. Draw a line from that string to the same finger on your outline. Do the same for every string, one at a time.

03 Does your hand work like the toy?



Trace each string to the finger it curls.

Look closer. Watch where the toy finger bends, then watch your own finger bend.

A little explanation. Both follow one plan: a pull, a string, and joints that bend. Your finger bends at three joints, mostly one way, like a door hinge. The muscles that curl your fingers sit mostly in your forearm, so you can feel them move there.

Try this. Curl your pointer finger slowly and count the places it bends. Curl the toy's pointer finger and compare. Rest your other hand on your forearm and wiggle your fingers. What do you feel move?

04 What opens the hand again?



Something has to pull the finger open again.

Look closer. Look at the back of the toy hand for elastic, a spring, or a second string.

A little explanation. A muscle can only pull, never push. So a second set of tendons runs along the back of each finger and pulls it straight. Many toys use an elastic cord or a spring instead. Some tendons on the back of your hand are linked, which is why one finger can be hard to lift alone.

Try this. Let go of a toy string and watch whether the finger springs back. Look for what pulls it open. Then lay your hand flat, fold your middle finger under, and gently try to lift your ring finger.